

Modeling *w76* Mutation Behavior Through Differential Heteroplasmy

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0. Biological Motivation

Development and behavior of the *w76* mutation in *Caenorhabditis elegans* remains largely uncharacterized. However, the mutation has been observed to influence mechanisms underpinning mitochondrial purifying selection by affecting levels of heteroplasmy when introduced to the *uaDf5* mitochondrial deletion. The following section discusses relevant known trends in *uaDf5* heteroplasmy during self-replication in the presence of the *w76* mutation.

0.1. Nature of Self-Replication in *C. elegans*

There are certain assumptions that are made about the self-replication process in *C. elegans* that informs our modeling of *w76* behavior.

Firstly, we assume that, for each self-replication event, the number of progeny laid by the individual will follow an approximately normal distribution:¹

$$X \sim N(\mu_X = 250, \sigma_X^2 = 200)$$

Then, note that the effect that *w76* has on *uaDf5* heteroplasmy is incremental across generations of self-replication. This means that, in the presence of the *w76* trait, each successive progeny retaining the trait will have an incrementally smaller proportion of *uaDf5*-affected mitochondria compared to its parent. This assumption will not be made for the naive model (**Section 1**), but will be accounted for in the state-aware model (**Section 3**).

Lastly, we assume that, after four generations of self-replication in the *w76* genetic background, purifying selection “completes” and wildtype homoplasmy is achieved (that is, the percentage *uaDf5*-affected mitochondria still present will be zero).

¹This assumption is actually not necessary/useful when only considering expected values. This assumption will be made, however, if the branching process is introduced, as it is a large part in characterizing the variance of the progenial distribution.

0.2. Dominant vs. Recessive

One important characteristic of any mutation in context of inheritance is whether it is dominant or recessive. It is currently unknown whether *w76* is dominant or recessive; however, we can deduce which it is by observing the mutation's effects on *uaDf5* heteroplasmy across rounds of self-replication.

For the models discussed here, expected results for both the dominant and recessive cases are calculated.

1. Naive Model of Propagation

It is useful to first develop a model for calculating the genotypic distribution of progeny in a population as a function of the starting population's genotypic distribution as well as the generational distance between the initial and "final" distribution. Conveniently, this can also give a rough boolean estimation of the proportion of progeny that has the ability to lose *uaDf5* at any given generation.

1.1. Naive Markov Transition Matrix

Suppose a vector $\vec{v}$ which represents the genotypic distribution of worms at generation g_0 . Then, the function $\vec{p}(\vec{v}, g_x) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ will represent the genotypic distribution of the progeny at generation g_x .

We first establish a Markov transition matrix that governs how different genotypes propagate across generations. For example, a homozygous parent will only ever self-replicate to create homozygous progeny, whereas a heterozygous parent will have a varied distribution of progeny. In order of $(\text{hom}^-, \text{het}, \text{hom}^+)$, the Markov matrix for this model will be:

$$M_N = \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0 & 1 \end{pmatrix}$$

Then, for some $\vec{v}$ of the same shape and order and $g_x \in \mathbb{N}$:

$$\vec{p}(\vec{v}, g_x) = \vec{v} \cdot M_N^{g_x}$$

Notice that both homozygous genotypes are "sinks": once an individual "enters" homozygosity, it is impossible for it to propagate anywhere else. This contrasts with the heterozygous genotype, which serves as a "generator" that pushes individuals into sinks.

As a quick test, we can look at the distribution of genotypes after $g_x = 4$ self-replication rounds. Suppose we have vector $\vec{v} = (0.5 \ 0.5 \ 0)^\top$ as our initial

distribution. Then,

$$\vec{p}(\vec{v}, 4) = \vec{v} \cdot M_N^4 = \begin{pmatrix} 0.5 \\ 0.5 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0 & 1 \end{pmatrix}^4 = \begin{pmatrix} 0.734375 \\ 0.03125 \\ 0.234375 \end{pmatrix}$$

Crucially, the sum of elements in $\vec{p}$ will always be equal to one. Notice again that the generator of heterozygous individuals trends towards the homozygous sinks.

2. Expected Distribution of Phenotypes

The above, for straightforward genotype-phenotype relationships, is sufficient to determine expected distributions. However, because *w76* behaves such that it will demonstrate incremental loss of *uaDf5*-affected mitochondria, we must consider the trait to be quantitative rather than boolean.

To inform a more accurate model, we should consider some emergent properties that stem from the combination of two “rules”:

1. *w76* incrementally decreases *uaDf5*-affected mitochondria with each successive generation, reaching homoplasmy at the fourth cycle of self-replication.
2. The generational distance between the initial and final populations will be finite, meaning there will be a non-zero proportion of heterozygotic individuals.

2.1. The *uaDf5* Expression Ladder

In both the dominant case, there will be a proportion of progeny at every generation that will “lock” itself from losing *uaDf5* heteroplasmy beyond its current state. This happens when lines fall into the hom^+ sink. However, whereas we previously grouped all $+/+$ individuals together under the genotype category, we cannot treat the phenotype the same way. This is because progeny that come from $+/-$ parents (e.g. at generation g_k) that become $+/+$ (at generation g_{k+1}) will lock its heteroplasmy level at the level of its parent. This means that, for two lines that enter $+/+$ at different generations, *uaDf5* heteroplasmy levels will generally be different as well.

For the recessive case, all $+/-$ will only be observed at the initial heteroplasmy level, since $+/-$ self-ing will never drop *uaDf5* levels; it will only produce $-/-$ that will drop in the next generation. However, $-/-$, trivially, will never produce progeny that enters $+/-$ or $+/+$.

Since every generation will have a proportion of its progeny enter $+/+$, beginning from the first self-replication cycle, we can expect to see a small number of lines that will always hold steady at the heteroplasmy level of their parents.

Conversely, heterozygous lines can fall into the $-/-$ sink as well; then, the line can never leave the *w76* background and therefore will always show incrementally decreasing heteroplasmy levels when compared to its parent.

In the dominant case, $+/-$ lines will move down with the $-/-$ lines, leaving $+/+$ “rungs” at each generation. However, in the recessive case, $+/-$ lines will stay at the original heteroplasmy level of its parent until lines enter $-/-$. Therefore, the top “rung” of the ladder (i.e. the point of greatest heteroplasmy, which should be at generation g_0) will contain all of the $+/-$ that have not lost any *uaDf5*-affected mitochondria.² At every generation, some of those initial $+/-$ lines will enter $-/-$ and begin reducing *uaDf5* levels. As such, for each generation, a single rung of the ladder (apart from the top rung) will contain only $-/-$, and the top rung containing all $+/-$ as well as any $+/+$ lines that existed in the initial genotypic distribution.

2.2. Finiteness of Generational Distance

The intuition behind the finiteness of generational distance is that, since $g_x \not\rightarrow \infty$, there will always be *w76*-background lines that do not reach homoplasmy. In this discussion, we will assume that there have been a total of four generations of self-replication ($g_x = 4$), just enough time for lines that have always had *w76* as a background to reach wildtype homoplasmy.

The dominant case is rather straightforward: there will be a large proportion of lines that reach wildtype homoplasmy ($\%hom^- + \%het$) and a smaller proportion distributed across g_x or $g_x + 1$ rungs made up of $+/+$ lines ($\%hom^+$).³

The recessive case is trickier: at each rung, the population displaying that level of heteroplasmy will be made up of only $-/-$ lines that are in the process of trending towards wildtype homoplasmy. Therefore, each successive rung (from bottom to top) represents one generation of delay for remaining $+/-$ lines in producing $-/-$ progeny.

3. State-Aware Model of Propagation

To represent the true phenotypic distribution that can be observed at generation g_x given genotypic distribution at generation g_0 , we must not only take into account not only what genotype a specific line is, but how long that line has been that genotype. To reiterate the properties governing our model:

1. Lines with *w76* will incrementally lose *uaDf5*-affected mitochondria in progeny.

²Since we assume that the generational distance between start and end is finite and reasonable, we can also assume that there will always be $+/-$ lines that still have not “fallen” from the starting rung.

³ $g_x + 1$ rungs if the initial population contained $+/+$ lines. There are g_x rungs that are caused by the self-replication cycles.

2. A line with four generations of self-replication in the *w76* background are considered to have reached wildtype homoplasmy.
3. Progeny count from any individual parent is assumed to be normally distributed according to the following:⁴

$$X \sim N(\mu_X = 250, \sigma_X^2 = 200)$$

3.1. *w76*'s Adjusted Markov Transition Matrix

Notice that the naive Markov transition matrix above does not change when considering the dominant or recessive backgrounds. This is because it tracks genotypes, not phenotypes. However, this is not true for the state-aware matrix (expanded in **Section 3.1.1**).

3.1.1. Dominant Case: The “Hypermatrix” Intuition

It is useful to first develop a heuristic about how we want to track this time signature for the dominant case.

Below is the definition for the matrix-of-matrices representing *w76* progenial behavior (from now referred to as the “hypermatrix”) we will use for the rest of the discussion.⁵

Consider the dominant form first.

Supposing the following matrix definitions:

$$M_N = \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0 & 1 \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad O = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Then, we can write the hypermatrix for a generational awareness up to 4 generations to be the following:

$$H^{(4)} = \begin{pmatrix} O & M_N & O & O & O \\ O & O & M_N & O & O \\ O & O & O & M_N & O \\ O & O & O & O & M_N \\ O & O & O & O & I \end{pmatrix}$$

The order of each submatrix will follow the original (hom^- , het , hom^+), and the hypermatrix is sorted in order of increasing generation (from 0 to 4).

⁴This assumption is actually not necessary/useful when only considering expected values. This assumption will be made, however, if the branching process is introduced, as it is a large part in characterizing the variance of the progenial distribution.

⁵Usually, the “hypermatrix” is actually called a block matrix, which segments a larger matrix into individual sub-matrices.

Notice first that, for generations 0 to 3, there is only one matrix to “go to”, and that they are diagonally distributed. This means that every progeny in each generation, if they are in generations $0 \leq g \leq 3$, will always move on to the “next” generation $g + 1$. This is important, because we want to know the time signature of all three genotypes: from **Section 2**, in the dominant case, heteroplasmy levels for all genotypes vary depending on what generation it is in.

We defined this hypermatrix to have a generational awareness of only up to four generations, because that is the amount of time required to reach wildtype homoplasmy for loss-phenotype lines. As such, at generation 4, we set the only possible transition matrix for all lines to take to the identity matrix (i.e. a sink), since we want them to stop propagating to the next generation.

The usable Markov transition matrix for the dominant case $M_{S,\text{dom}}$ is simply the result of substituting matrix variables for their matrix representations (not shown here).

3.1.2. Recessive Case: Direct Representation

The recessive case, conveniently, does not have to consider . From **Section 2**, $+/+$ and $+/-$ lines never vary in heteroplasmy level, and therefore do not have to be tracked with a time signature. This reduces the matrix size and complexity much more, since we now only need to track one genotype ($-/-$) across time.

The Markov transition matrix for the recessive case is defined as follows:

$$M_{S,\text{rec}} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0.25 & 0 & 0 & 0 & 0.5 & 0.25 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Note that this matrix, in contrast to all the ones before, will follow the order of $(\text{hom}_0^-, \text{hom}_1^-, \text{hom}_2^-, \text{hom}_3^-, \text{hom}_4^-, \text{het}, \text{hom}^+)$.

3.2. Example Usage

We verified these derivations for the new transition matrix using Python 3.14 and the NumPy library. `##` 3.3. Future Directions: Multitype Branching Process